SPECIFICATION

1 Background of the Invention

Over-the-horizon communication has traditionally relied on either bouncing signals off the ionosphere, using dedicated communication satellites, or laying extensive terrestrial infrastructure like fiber optic cables. Dedicated satellites are expensive to build and launch, and their communication channels are often monitored and controlled. Terrestrial infrastructure is vulnerable to physical disruption and is not available in many remote or oceanic areas.

While techniques like tropospheric scatter exist for point-to-point military communication, they are limited in range and require specific atmospheric conditions. There exists a need for a communication method that can bridge long distances without relying on dedicated, active satellite relays or vulnerable ground infrastructure.

The recent proliferation of thousands of satellites in Low Earth Orbit (LEO), such as those in the Starlink, OneWeb, and Kuiper constellations, has filled the near-Earth space with a dense and predictable cloud of metallic objects. While these satellites are designed for active communication with their proprietary ground stations, their physical bodies (e.g., chassis, solar panels) are also reflective to radio frequencies. To date, these objects have not been systematically utilized as passive reflectors for unrelated, third-party communication links. The technical challenge lies in the extreme complexity of predicting the precise trajectory of thousands of rapidly moving objects and calculating the exact moment when one will be perfectly positioned to create a specular reflection between two specific points on Earth.

2 Summary of the Invention

The present invention provides a method and system to overcome these limitations by treating the existing cloud of orbiting objects as a dynamic, statistically predictable field of potential signal reflectors.

A primary object of the invention is to provide a system that calculates a spatio-temporal intersection point—a specific point in space at a specific future time—where an orbiting object will be positioned to reflect a signal from a transmitter to a receiver.

Another object of the invention is to provide a method for retrieving real-time or near-real-time orbital parameter data (e.g., Two-Line Element sets) for a large number of orbiting objects.

A further object of the invention is to provide a computational engine that predicts the future trajectories of said objects based on this data and identifies a candidate object and an associated event time for a communication link.

The system comprises a transmitting station, a receiving station, and a central computation engine. The computation engine continuously ingests orbital data, calculates potential reflection geometries, and provides the transmitting station with precise firing solutions (i.e., target azimuth, elevation, and timing) to establish a link.

In one embodiment, the transmitting and receiving stations are equipped with movable, steerable antennas. In another embodiment, the stations are equipped with a plurality of fixed antennas pointing towards cardinal directions, and the system waits for an orbiting object to align with one of the pre-defined transmission paths.

The invention enables a form of communication that is highly secure, as the specific satellite and timing used for any given transmission are ephemeral and selected from thousands of possibilities, making interception extraordinarily difficult. It also provides a potential

communication method for remote areas, disaster zones, or any scenario where traditional infrastructure is unavailable.

3 Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the overall system, showing a ground transmitter, a target receiver, and a non-cooperative satellite acting as a passive reflector.

FIG. 2 is a geometric diagram illustrating the reflection principle, showing the incident and reflected signal paths and the object's position relative to the transmitter and receiver.

FIG. 3 is a flowchart of the computational method for identifying a viable communication opportunity.

FIG. 4 is a diagram contrasting the operational modes for a system with a movable antenna versus a system with fixed cardinal antennas.

4 Detailed Description of the Invention

The core of the invention is a computational system that solves a complex, four-dimensional (3D space + time) kinematic problem. The system is designed to find a future moment in time when a non-cooperative orbiting object will pass through a specific point in space that satisfies the geometric conditions for specular reflection between a ground transmitter and a ground receiver.

4.1 System Architecture

The system consists of three main components:

1. **Data Ingestion Module:** This module is responsible for periodically fetching orbital parameter data for a large plurality of orbiting objects. A primary source for this data is the public Two-Line Element (TLE) set repository maintained by CelesTrak. This data provides the necessary parameters to model the orbit of each object. The data is cached locally to minimize external dependencies.

2. **Computational Engine:** This is the central processing unit of the system. It takes as input the locations of the sender and receiver and the orbital data from the ingestion module. It performs a search over a future time window to find a viable communication opportunity. Due to the computational intensity, the engine is preferably implemented as a distributed, massively parallel system. A cloud-based task queue architecture is used, where hundreds or thousands of small computational tasks are created, each responsible for evaluating a specific, narrow slice of the future time window.

3. **Transceiver Stations:** The physical ground stations. Each station includes a directional antenna (or array of antennas) and a radio transceiver. The transmitting station is controlled by the computational engine, which dictates the precise moment to send a directed signal.

4.2 Computational Method

The computational method differs based on the type of antenna system employed.

4.2.1 Movable Antenna Mode

This mode offers the highest flexibility and is suitable for dynamic or mobile communication needs (e.g., ship-to-shore).

1. **Initialization:** The system receives the geographic coordinates (latitude, longitude) of the sender (S) and receiver (R).

2. **Time-Slicing:** The system defines a future search window (e.g., the next 10 minutes) and divides it into discrete time steps, t_i (e.g., every 30 seconds). The following steps are performed for each t_i.

3. **Ephemeris Calculation:** Using a celestial mechanics library such as Skyfield, the system calculates the Earth-Centered, Earth-Fixed (ECEF) position vectors for the sender, $P_S(t_i)$, and the receiver, $P_R(t_i)$.

4. **Satellite Position Prediction:** For each of the thousands of satellites in the TLE dataset, the system predicts its ECEF position vector, $P_{sat}(t_i)$.

5. **Geometric Reflection Check:** For a signal to reflect from the satellite to the receiver, the satellite's position must satisfy the law of reflection. The system models the Earth's surface at the satellite's ground track as the reflection plane. It calculates the surface normal vector, N_{sat}, at the satellite's position. The vector from the sender to the satellite, $V_{in} = P_S - P_{sat}$, and the vector from the receiver to the satellite, $V_{out} = P_R - P_{sat}$, are calculated. A valid reflection occurs if the angle of incidence equals the angle of reflection with respect to the normal vector. This is computationally verified by checking if:

$$\frac{V_{in} \cdot N_{sat}}{\|V_{in}\|} \approx \frac{V_{out} \cdot N_{sat}}{\|V_{out}\|}$$

within a predefined tolerance.

6. **Line-of-Sight Check:** The system verifies that the satellite is above a minimum elevation angle (e.g., 10 degrees) from the horizon for both the sender and the receiver to ensure a clear line of sight.

7. **Firing Solution:** If a satellite and time step (sat_j, t_i) satisfy these conditions, a firing solution is generated. The system outputs the satellite's name, the precise time of the event, and the required azimuth and elevation for the sender's antenna to target the satellite at that instant. A countdown is initiated for the transmission.

A key optimization involves filtering the satellites based on distance and kinematics. For transmissions between different hemispheres, only satellites moving in the correct direction (i.e., ascending or descending orbit) are considered.

4.2.2 Fixed Cardinal Antenna Mode

This mode is suitable for communication between fixed installations, where movable antennas are impractical. Each station is assumed to have an array of antennas pointing to fixed cardinal directions (e.g., North, East, South, West) at a fixed elevation.

1. **Initialization:** The system receives the sender and receiver coordinates and pre-calculates the fixed line-of-sight (LOS) vectors for each antenna in the ECEF coordinate system.

2. **Plausible Path Pruning:** The system determines which pairs of sender-receiver antennas are geometrically plausible. For example, for a sender in Texas to communicate with a receiver in New York, a plausible path might involve the sender's North-facing antenna and the receiver's South-facing antenna. This pruning dramatically reduces the search space.

3. **Keyhole Search:** For each plausible antenna pair and for each time step t_i, the system iterates through the satellites. It checks if a satellite's predicted position $P_{sat}(t_i)$ falls within a narrow conical "keyhole" defined by the sender's antenna LOS vector.

4. **Reflection Vector Calculation:** If a satellite is found in the sender's keyhole, the system calculates the ideal reflection vector based on the surface normal at the satellite's position (N_{sat}) and the incoming signal vector (which is aligned with the sender's LOS).

5. **Recipient Alignment Check:** The system then checks if this calculated reflection vector aligns with the actual vector from the satellite to the receiver ($P_R - P_{sat}$). It also verifies that this vector falls within the recipient antenna's corresponding keyhole.

6. **Solution Generation:** If all conditions are met, a communication opportunity is confirmed. Since the antennas are fixed, the firing solution only consists of the precise transmission time.

4.3 System Implementation and Embodiment

A prototype of the system has been implemented as a distributed web application using the Python programming language, the Flask web framework, and the Skyfield library. The computational engine is deployed on a cloud platform (e.g., Google App Engine) and uses a task queue (e.g., Google Cloud Tasks) to parallelize the search across hundreds of virtual workers. A NoSQL database (e.g., Google Firestore) is used to manage the state of the search and store the results.

The system intelligently selects which satellite constellations to search based on the Haversine distance between the two ground points. For distances greater than a threshold (e.g., 5000 km), higher-altitude MEO and GEO satellites are prioritized, as LEO satellites would be below the horizon. For shorter distances, the denser LEO constellations are used.

This embodiment demonstrates that the complex calculations required to find a viable bounce opportunity can be performed within a practical timeframe (typically under 30 seconds), making the system feasible for real-world application. To further enable one of ordinary skill in the art to practice the invention, a complete, working source code implementation of this embodiment is provided in Appendix A.